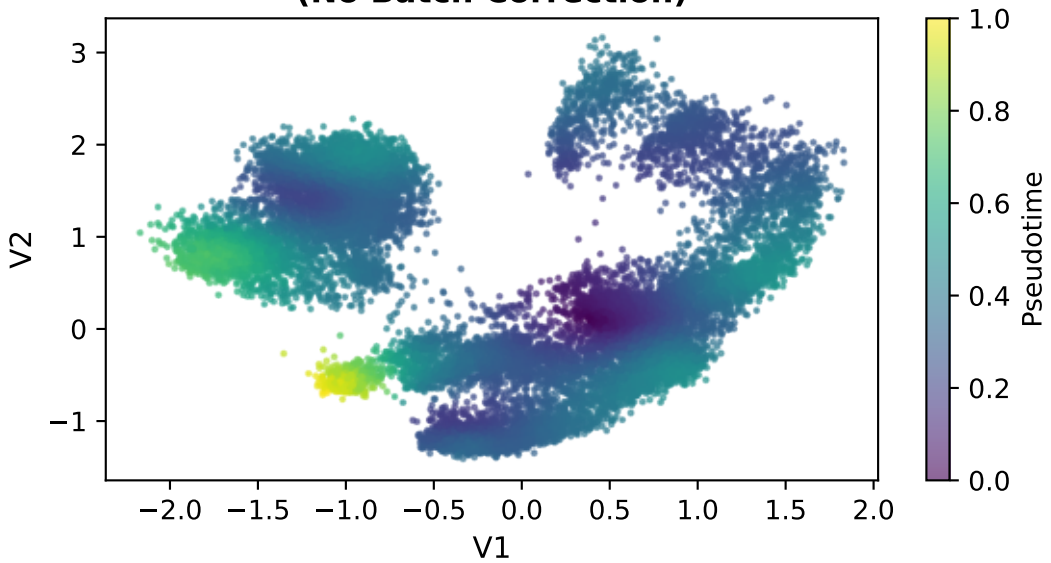
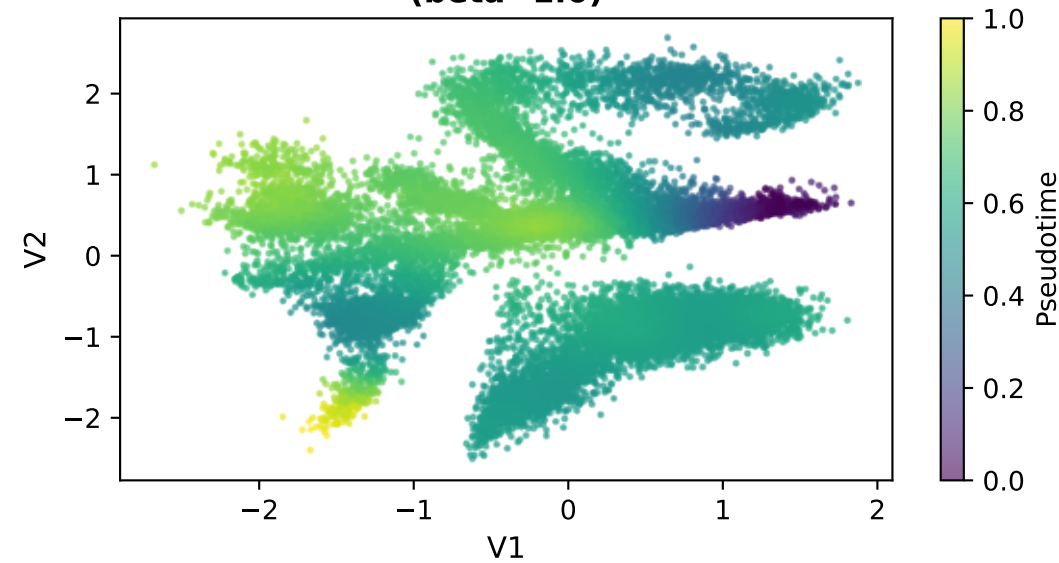


V Space Comparison: Concatenation vs Attention-Based Batch Correction (beta=1.0)

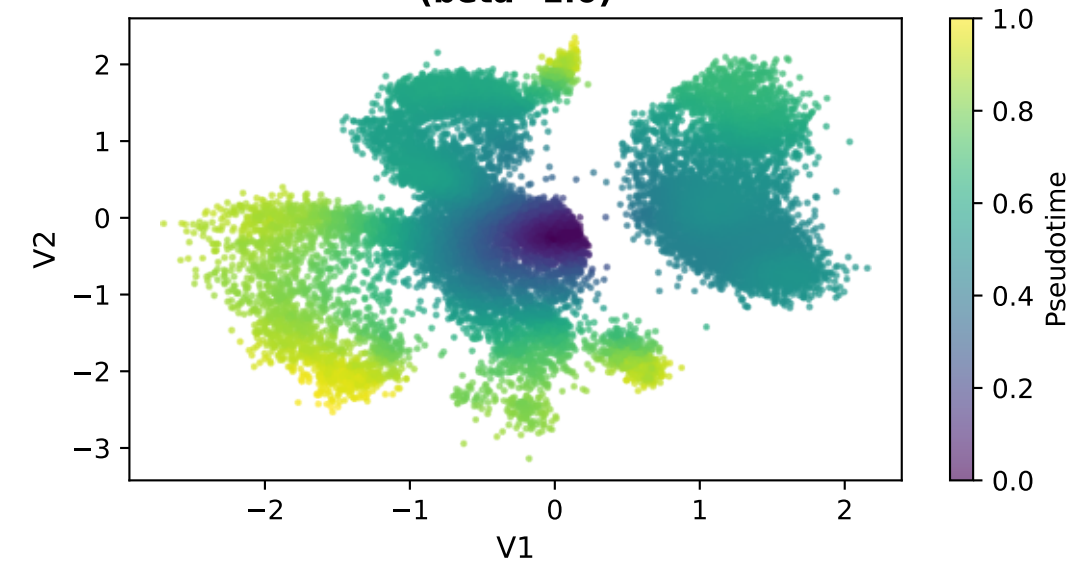
**Native Decipher
(No Batch Correction)**



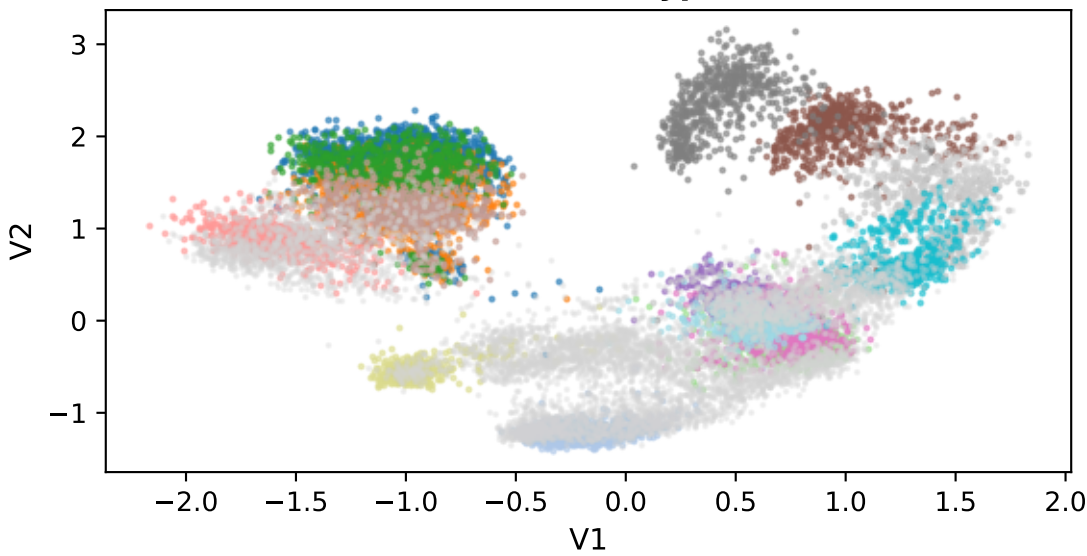
**Concatenation
(beta=1.0)**



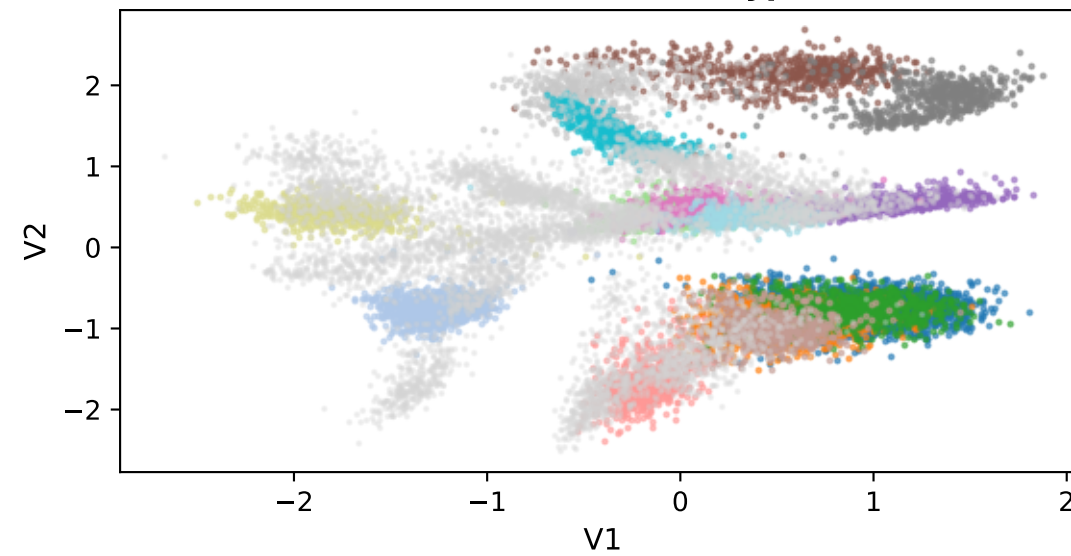
**Attention (2 heads)
(beta=1.0)**



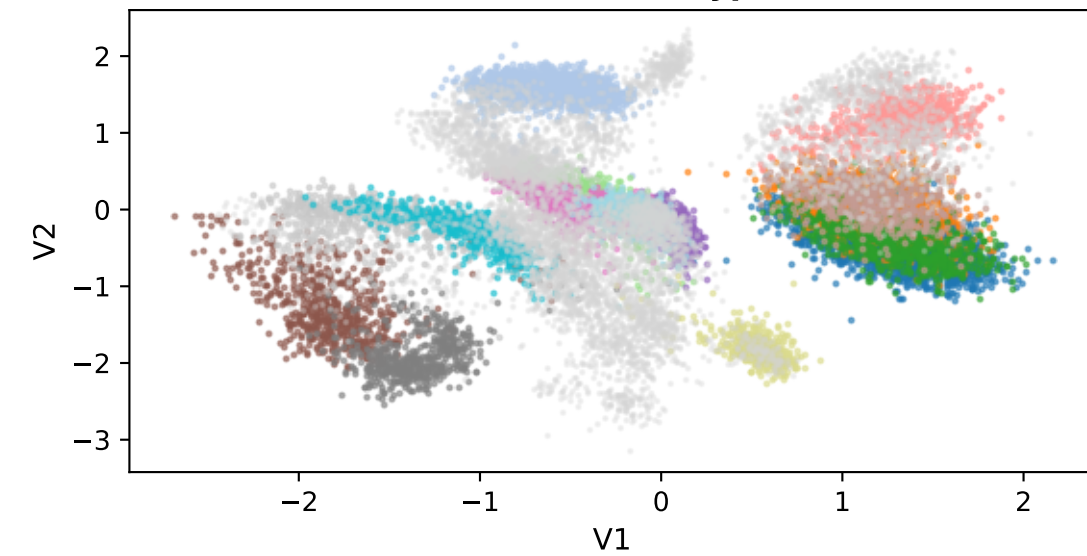
Native (Cell Types)



Concatenation (Cell Types)

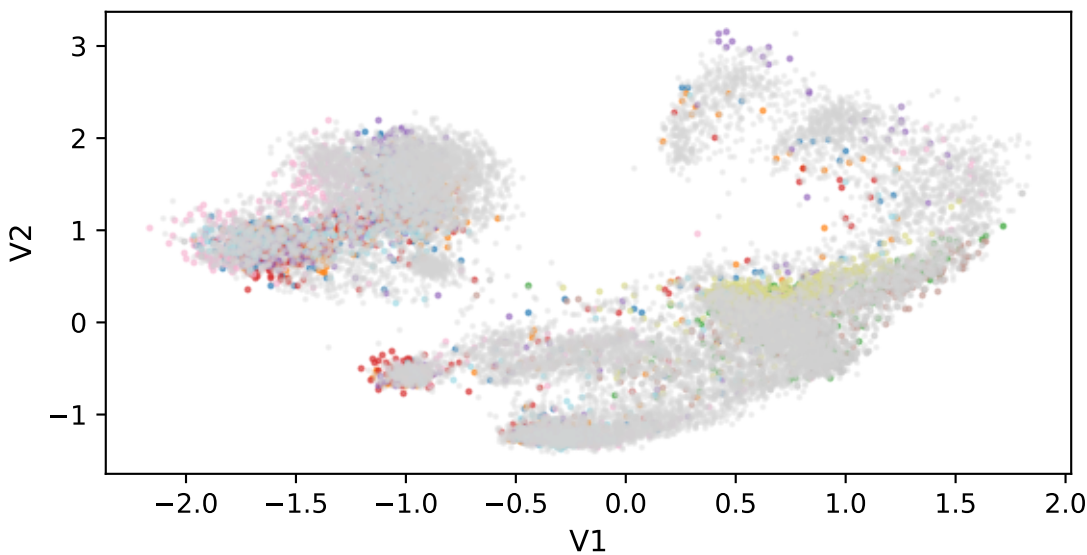


Attention (Cell Types)

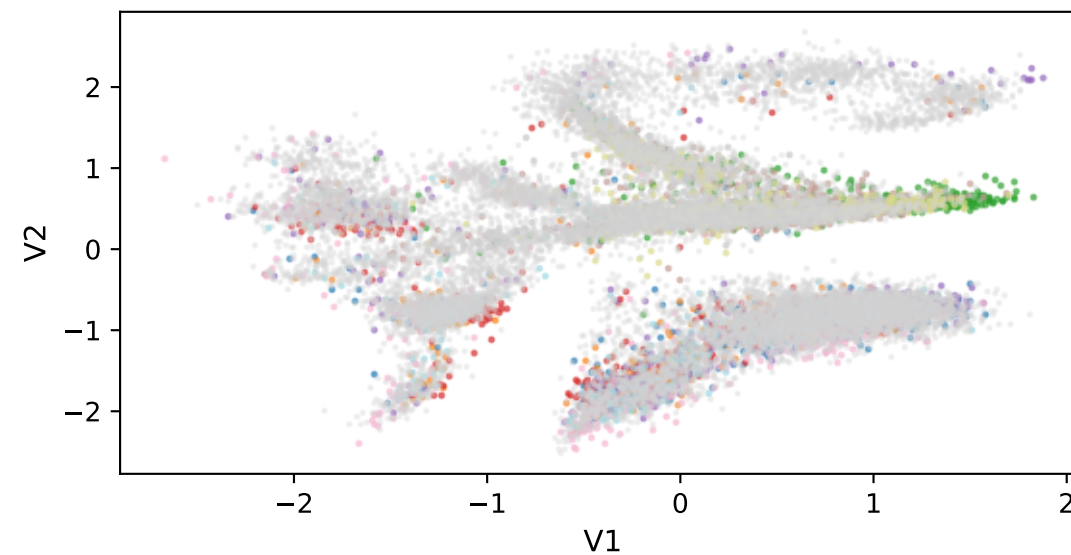


- CD4 Naive
- CD14 Mono
- CD4 Central Memory
- CD8 Naive
- Early GMP
- CD8 Effector Memory
- HSC
- Polychromatic Erythr
- CD4 Effector Memory
- Cycling Progenitor
- Orthochromatic Eryth
- Pro-Erythroblast
- Mature B
- CFU-E
- LMPP
- Other types

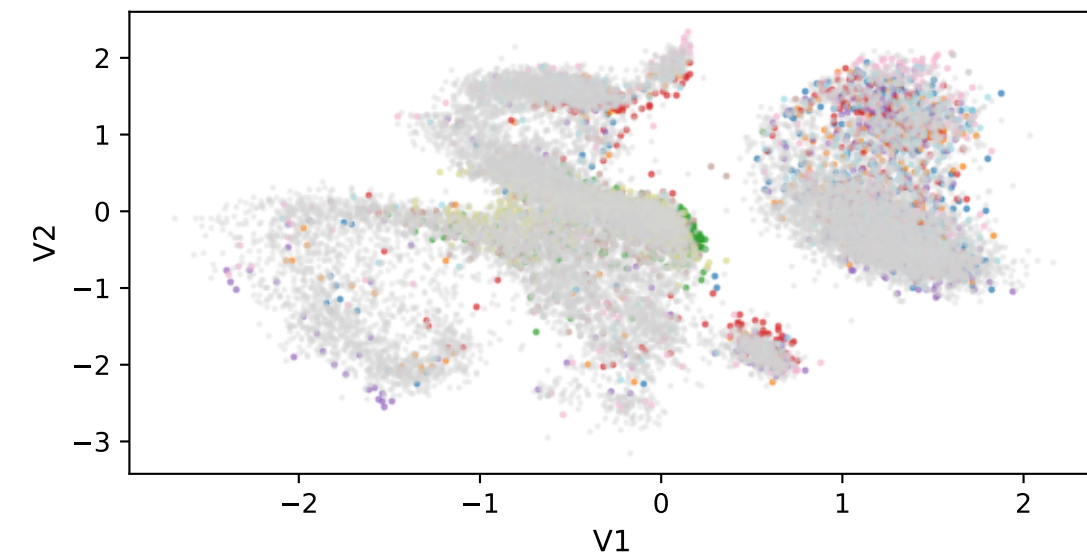
Native (Donors)



Concat (Donors)

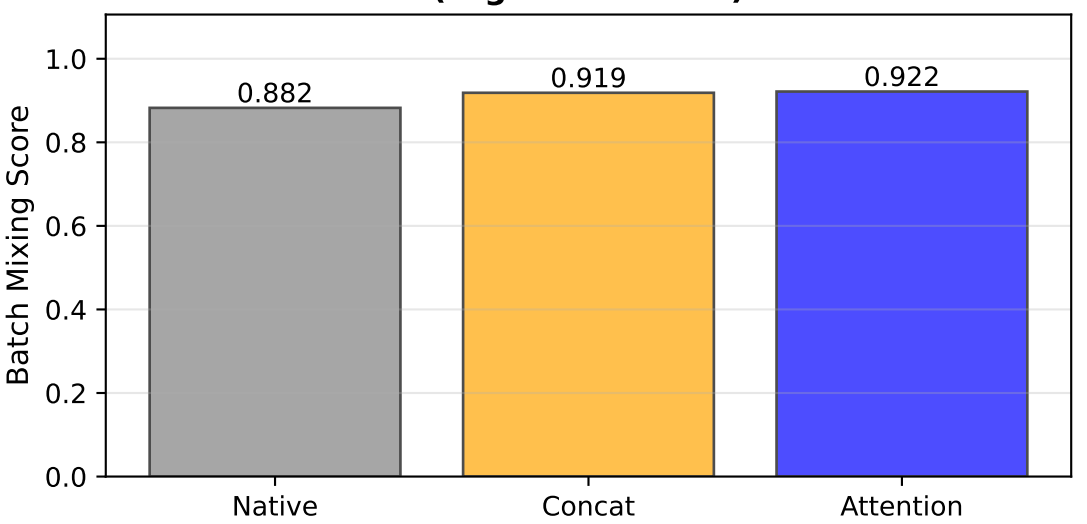


Attention (Donors)

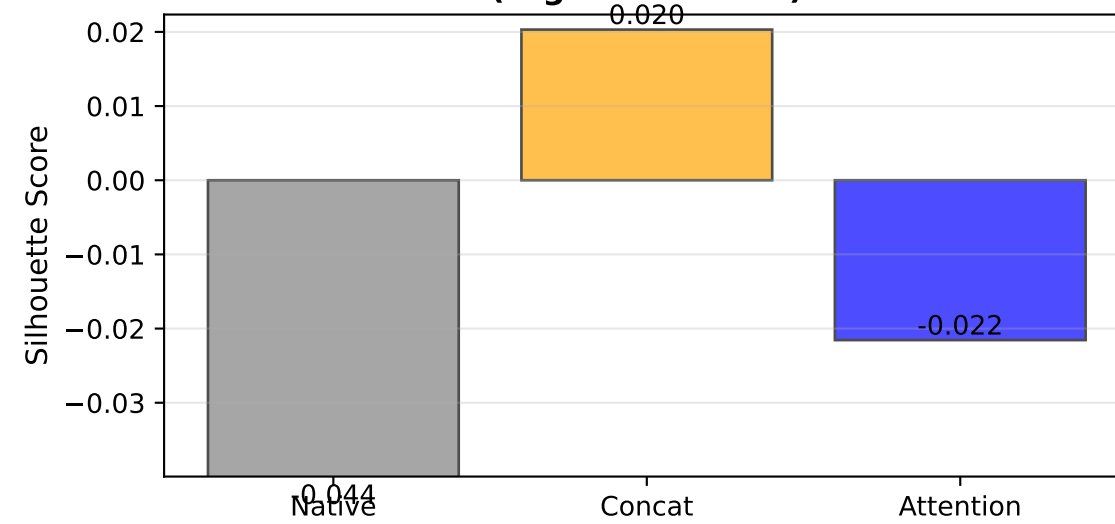


- Oetjen2018_K
- Oetjen2018_N
- Ainciburu2022_e
- Oetjen2018_Sk
- Oetjen2018_H
- Ainciburu2022_e
- HCA_BM_BM2
- Ainciburu2022_y
- Mende2022_BM_DO
- Oetjen2018_O
- Other donors

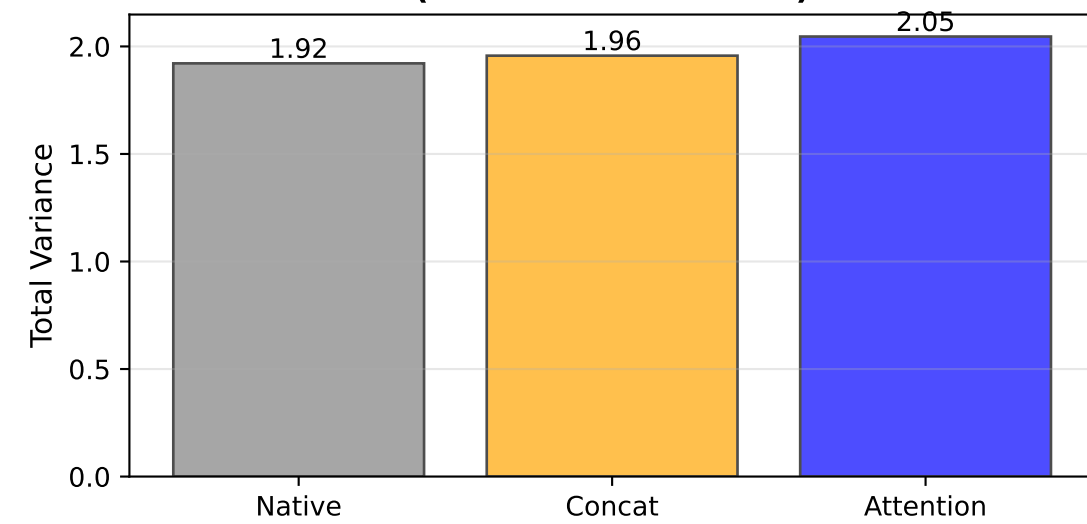
**Batch Mixing Quality
(Higher = Better)**



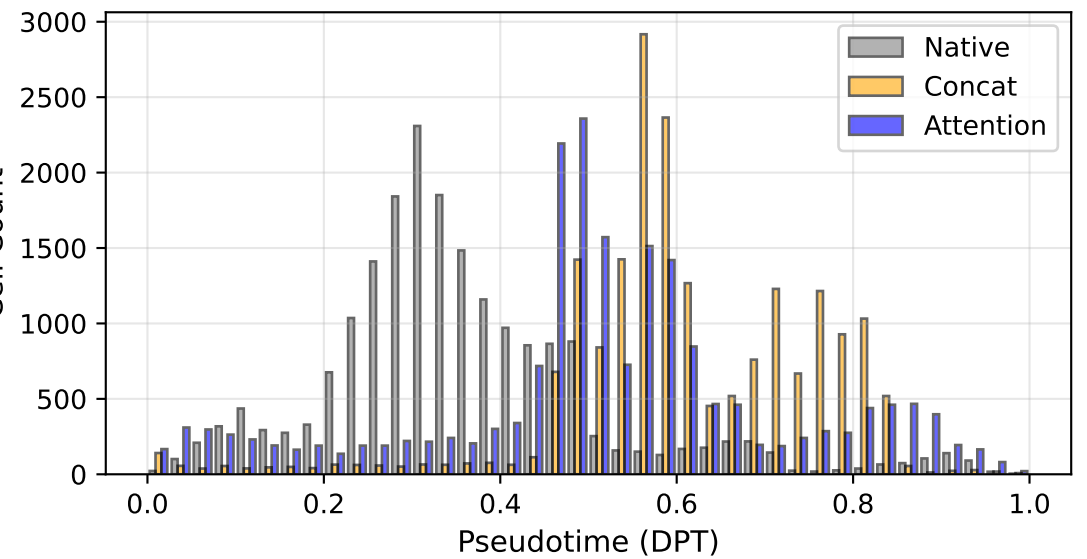
**Cell Type Separation
(Higher = Better)**



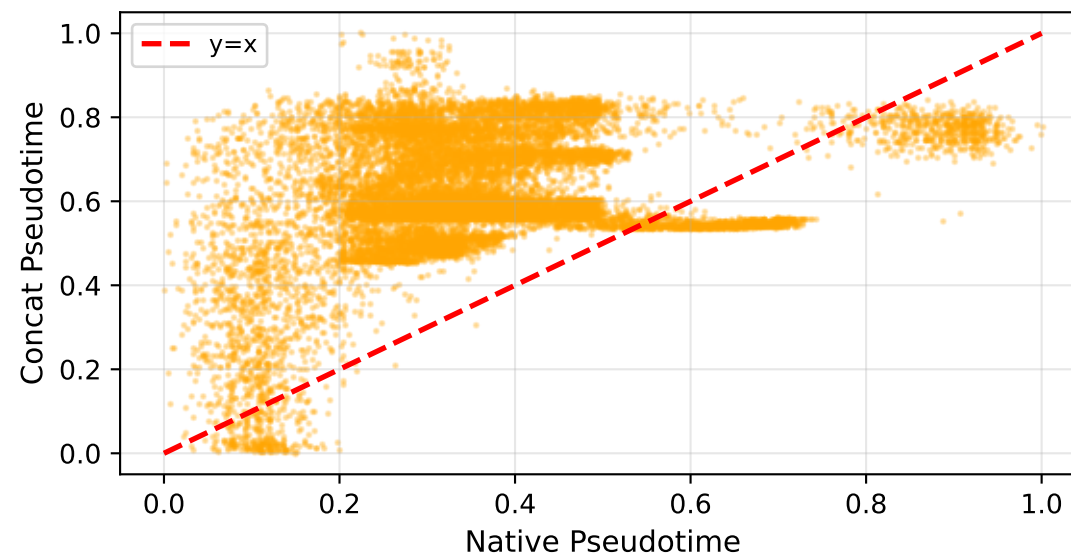
**V Space Variance
(Information Content)**



Pseudotime Distribution



**Concat vs Native
Corr: 0.342**



**Attention vs Native
Corr: 0.492**

